Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence

Understanding Linear Independence in Vector Spaces
Reading • 18 min

Linear Independence, Basis, and Rank
Ungraded App Item • 20 min

Linear Independence
Reading • 30 min

Linear Mapping

Exploring Transformations: Understanding Linear Mappings
Reading • 1h

Linear Mapping: Part 1
Video • 8 min

Linear Mapping: Part 2
Video • 8 min

Linear Mapping
Reading • 5 min

Matrix Representation
Reading • 6 min

Practice Quiz: Linear Mapping
Graded Assignment • 18 min

Lesson 3: Eigenvalues and Eigenvectors

Introduction to Eigenvalues and Eigenvectors
Reading • 40 min

Eigenvectors and Eigenvalues: Examples and Applications
Reading • 30 min

Lesson 4: Cholesky Decomposition

Cholesky Decomposition
Reading • 5 min

Solved Examples
Reading • 10 min

Lesson 5: Singular Value Decomposition

Introduction to Singular Value Decomposition (SVD)
Reading • 28 min

Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors
Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Matrix Representation

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Matrix Representation of Linear Mappings

A linear mapping (or linear transformation) can be effectively represented using matrices. This allows us to use matrix algebra to perform transformations, making computations more efficient and easier to understand.

Basics of Matrix Representation

Given a linear mapping $\Phi: \mathbb{R}^n \to \mathbb{R}^m$, there exists an $m \times n$ matrix A such that for any vector $x \in \mathbb{R}^n$:

$$\Phi(x) = Ax$$

Here's how the matrix representation works:

Mapping Basis Vectors:

- Consider the standard basis vectors $e_1, e_2, \ldots e_n$ in $\mathbb{R}^n$. These vectors have 1 in one coordinate and 0 in all others.
- The action of the linear mapping Φ on these basis vectors can be used to construct the matrix A.

Constructing the Matrix:

- Apply Φ to each basis vector:

$$\Phi(e_1) = Ae_1, \quad \Phi(e_2) = Ae_2, \quad \ldots, \quad \Phi(e_n) = Ae_n$$

- Each resulting vector $\Phi(e_i)$ is a column in the matrix A.

Forming the Matrix:

- If $\Phi(e_i)$ results in the vector x_i in $\mathbb{R}^m$, then A is formed by placing x_i as the ith column of A.

$$A = [\Phi(e_1)\, \Phi(e_2), \quad \ldots, \quad \Phi(e_n)]$$

Example

Consider a linear mapping $\Phi: \mathbb{R}^2 \to \mathbb{R}^2$ defined by:

$$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ 3x_1 + 4x_2 \end{pmatrix}$$

Apply Φ to Basis Vectors:

- Basis vectors in $\mathbb{R}^2$ are $e_1 = (\begin{bmatrix} 1 \\ 0 \end{bmatrix})$ and $e_2 = (\begin{bmatrix} 0 \\ 1 \end{bmatrix})$
- Apply Φ to e_1:

$$\Phi(e_1) = \Phi\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \cdot 1 + 0 \\ 3 \cdot 1 + 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

- Apply Φ to e_2:

$$\Phi(e_2) = \Phi\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 + 1 \\ 0 + 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

Form the Matrix:

- Place the results as columns in matrix A:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$$

Verify the Representation:

- For any vector $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ 3x_1 + 4x_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = Ax$$

Go to next item ✓ Completed

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